

VARSHA N

- +91 9585599344
- varshanandakumar069@gmail.com
- github.com/6-varsha-9
- linkedin.com/in/varsha-nandakumar-7713b825a

CAREER PROFILE

AI & Data Science undergraduate passionate about building intelligent, data-driven solutions through Machine Learning and AI. Strong analytical thinker with problem-solving skills, adaptability, and a commitment to continuous innovation. Quick learner and collaborative team player focused on creating impactful real-world solutions.

EDUCATION

B Tech Artificial Intelligence and Data Science

Dr.N.G.P.Institute of Technology, Coimbatore

GRADUATION - MAY 2026

CGPA - **7.89**

Higher Secondary (HSC)

Vellalar Matriculation Higher Secondary School, Erode

JUNE 2021 - MAY 2022

Percent - **84%**

Secondary School Leaving Certificate(SSLC)

Vellalar Matriculation Higher Secondary School, Erode

JUNE 2019 - MARCH 2020

Percent - **72%**

TECHNICAL QUALIFICATIONS

- Data Science & Analytics:** NumPy, Pandas, Data Visualisation, Power BI
- Deployment & Tools:** Flask, Docker, Jupyter Notebook, Google Colab
- Computer Vision:** OpenCV, MediaPipe, Image Processing, CNN
- Programming Skills:** Python, SQL, Java
- Machine Learning:** Scikit-learn, TensorFlow, Keras, Deep Learning, Model Evaluation, Feature Engineering
- Natural Language Processing:** NLTK, spaCy, Text Processing

ACADEMIC PROJECTS

Multi-Modal Deep Learning Framework for Autism Spectrum Disorder Screening

SEP 2025 - FEB 2026

- Developed a multi-modal deep learning system for early autism detection using facial expressions, body pose, and eye gaze from videos.
- Implemented CNN-Transformer, ST-GCN, and BiGRU models to extract features and improve prediction accuracy, with data preprocessing, training, testing, and evaluation.
- Applied Grad-CAM and SHAP to provide explainable AI predictions, making the model's decisions easier to understand.

AI-Powered Real-Time Sign Language Translator

AUGUST 2024 - MAY 2025

- Engineered a real-time ISL-to-text & speech system using CNN, Mediapipe, and OpenCV, classifying 27 gestures.
- Integrated gTTS & pyttsx3 for both online and offline speech synthesis.
- Deployed via Flask for browser-based real-time interaction.

Pneumonia Detection using Machine Learning

NOV 2023 - AUGUST 2024

- Developed and deployed a CNN-based chest X-ray classifier with 85% accuracy using TensorFlow, Keras, and Flask for web-based pneumonia detection.
- Performed image preprocessing, model training, testing, and evaluation to improve the performance and reliability of the classification model.
- Focused on building a scalable AI solution for healthcare applications, with potential use in clinical decision support and insurance claim assessment.

INTERNSHIPS

PYTHON DJANGO DEVELOPER INTERN

JULY - 2024

PINESPHERE SOLUTIONS PRIVATE LIMITED, COIMBATORE

- Assisted in developing and deploying cloud-based web applications for scalable access.
- Optimized MySQL and MongoDB databases, achieving a 15% reduction in query time.
- Enhanced responsive UI/UX design, improving user engagement during testing.
- Collaborated on full-stack development, integrating front-end and back-end features efficiently.

JAVA DEVELOPER INTERN

CODSOFT - ONLINE

APRIL - 2024

- Built mini-projects applying OOP principles (inheritance, file handling, exception management).
- Debugged and optimized programs, improving execution efficiency by ~20%.
- Strengthened problem-solving & backend development skills.

HONORS AND RECOGNITIONS WORKSHOPS

- Python Full Stack – Gained insights into modern app development & deployment.
- Ethical Hacking – Hands-on with information security fundamentals.

HACKATHONS

- Smart India Hackathon (Sep 2023) – Developed a Women Safety Drone Project.
- IoT & Automation Hackathon – Worked on automation-based projects with real-time applications.
- 24-Hour Hackathon, Dr. N.G.P. Institute of Technology (Jul 2023).

ACHIEVEMENTS

- Secured 2nd & 3rd prizes in Paper Presentations on AI/ML innovations.

CERTIFICATIONS

- Python for Data Science - **Great Learning**
 - Introduction to Artificial Intelligence - **Great Learning**
 - Introduction to Deep Learning - **Great Learning**
 - Certified in CyberSecurity - **ISC2** 
 - Power BI for Beginners - **Simplilearn**
 - Generative AI for Data Analysts - **Coursera** 
 - Data Fundamentals - **IBM** 
 - Get Started Building with Power BI - **Microsoft** 
 - Google Analytics Certification - **Google** 
 - AI-Powered Performance Ads Certification - **Google** 
 - Advanced IT Skills in Artificial Intelligence – **IBM SkillsBuild** 
-